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Programme : BACHELOR OF INFORMATION TECHNOLOGY (HONS) IN SOFTWARE ENGINEERING

Course Code & Name : IGB20303 PROBABILITY AND STATISTICS FOR IT

Course Lecturer(s) : SITI ZAHARIAH BINTI ABDUL WAHAB

Submission Title : FINAL PROJECT - PORBABILITY AND STATISTICS FOR IT

Deadline : Day 24 Month JUNE Year 2024 Time 12.30pm

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25/6/2024	Muhammad Amir Imran Bin Abdul Razak Khaizuran Amshar Bin Mohd Tarmizi Muhammad Arif Fahmi Bin Amaludin Chan Boon Hong Khairul Aqib Bin Jazrul Fuad	52213123124 52213123189 52213123164 52213123434 52213123225

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Investigating the Impact of Caffeine Consumption on Sleep Efficiency

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Abstract

This study was conducted to examine the relationship between human caffeine consumption and sleep efficiency. Another reason that this case study is being conducted is to determine whether reducing caffeine intake improves sleep efficiency or otherwise. Descriptive testing, including standard deviation and mean is utilized alongside hypothesis testing, T-testing and correlation analysis. Specifically, the Pearson Correlation Coefficient is applied to assess how caffeine affects human's sleep efficiency. The findings reveal that caffeine does not significantly impact sleep efficiency. This suggests that caffeine is not the primary factor causing decreased sleep efficiency. Instead, other factors such as environmental influences also play a significant role. Therefore, while caffeine consumption may affect sleep efficiency, it is not the only factor. Other variables must be considered to fully understand what affects sleep efficiency. The study shows that sleep efficiency is complex and needs a broader perspective when looking at sleep quality and its causes. This comprehensive approach is essential for developing effective strategies to improve sleep quality in individuals.

Keywords : (Caffeine Consumption; Sleep Efficiency; Pearson Correlation Coefficient)

1. Introduction

The primary issue to be investigated is the impact of caffeine consumption on sleep. Specifically, this study aims to examine how the amount of caffeine consumed per day affects an individual's sleep efficiency. By analyzing the relationship between caffeine intake and sleep patterns, the goal is to determine whether a reduction in caffeine consumption can lead to improved sleep efficiency. This investigation seeks to provide insights into whether consuming less caffeine could potentially enhance the overall quality of sleep. According to research by Carissa Gardiner and Jonathan Weakley, as cited on ScienceDirect, "The consumption of caffeine impairs subsequent total sleep time, sleep onset latency, wake after sleep onset, sleep efficiency, and sleep architecture." which shows that amount of Caffeine consumption negatively impacts sleep quality, particularly total sleep time, with larger doses. The method that we use to summarize the caffeine consumption data is by finding the mean and the standard deviation of the data. We also used One sample T test to test whether the mean average consumption is equal to 400 mg as it is the recommended amount of caffeine consumption per day. Lastly, we use Pearson Correlation Coefficient to find the relation or correlation between caffeine consumption and sleep efficiency.

2. Materials and Methods (Methodology)

2.1 Data Analysis

The analysis of the Sleeping Efficiency dataset, which includes attributes for sleep efficiency and caffeine consumption, involved both descriptive statistics and inferential statistics to test hypotheses and explore correlations.

2.2 Descriptive Statistics

We used mean and standard deviation because we wanted to summarize the central tendency and variability of the sleep efficiency and caffeine consumption data. This dataset aims to understand sleep patterns and factors affecting sleep quality, making it crucial to establish average values (mean) and assess the variability (standard deviation) within the data.

Mean: Calculating the mean provides us with the average sleep efficiency and caffeine consumption, offering a baseline for typical behavior in the dataset. This helps in identifying whether individuals, on average, have efficient sleep and how much caffeine they consume.

Standard Deviation: The standard deviation indicates the spread of the data points around the mean, providing insights into the variability of sleep efficiency and caffeine consumption. High

variability in sleep efficiency might suggest inconsistencies in sleep patterns, while variability in caffeine consumption can indicate different consumption habits among individuals.

- **Mean:** The arithmetic mean was calculated for both sleep efficiency and caffeine consumption. The formula used for the mean is:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

where $\bar{x}$ is the mean, x_i represents each data point, and n is the total number of data points.

- **Standard Deviation:** The standard deviation was calculated to measure the dispersion of the data around the mean. The formula used for the standard deviation was:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

where s is the standard deviation, x_i represents each data point, $\bar{x}$ is the mean, and n is the total number of data points.

2.3 Hypothesis Test

We used a one-sample t-test because we wanted to determine if the sample mean of sleep efficiency and caffeine consumption significantly differed from known or hypothesized population means. This is essential for understanding whether the observed sample represents the general population or if there are significant differences.

One-Sample T-Test: This test helps us determine if the average sleep efficiency and caffeine consumption in our sample are significantly different from a known population mean or a hypothesized value. For example, we might hypothesize that the average sleep efficiency should be 0.80 or that the average caffeine consumption should be 20 mg. The one-sample t-test allows us to statistically test these hypotheses.

The formula used for the one-sample t-test was:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

where t is the t-statistic, $\bar{x}$ is the sample mean, μ is the population mean, s is the standard deviation, and n is the sample size.

The t-test was performed separately for sleep efficiency and caffeine consumption to test specific hypotheses regarding their population means.

2.4 Correlation

We used Pearson Correlation Coefficient because we wanted to examine the relationship between sleep efficiency and caffeine consumption. This correlation helped us determine whether there was a statistically significant linear relationship between the two variables, which can provide insights into how changes in caffeine consumption might be related to changes in sleep efficiency. This analysis is crucial for understanding how lifestyle factors, such as caffeine intake, might influence sleep quality.

Pearson Correlation Coefficient: This measure helps us determine whether there is a statistically significant linear relationship between sleep efficiency and caffeine consumption. By calculating the correlation coefficient, we can assess whether higher caffeine consumption is associated with lower sleep efficiency or vice versa. This insight is valuable for identifying potential behavioral or physiological connections between caffeine intake and sleep quality.

The formula used for Pearson Correlation Coefficient was:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

where r is the correlation coefficient, x_i and y_i are the individual sample points for sleep efficiency and caffeine consumption respectively, $\bar{x}$ is the mean of sleep efficiency, and $\bar{y}$ is the mean of caffeine consumption.

This analysis helped determine whether there was a statistically significant relationship between the two variables, providing insights into how caffeine consumption might impact sleep efficiency.

3. Results and Discussion

1. Figures
2. Charts

Describe All Figures, Tables, Charts, Results (4 Results)

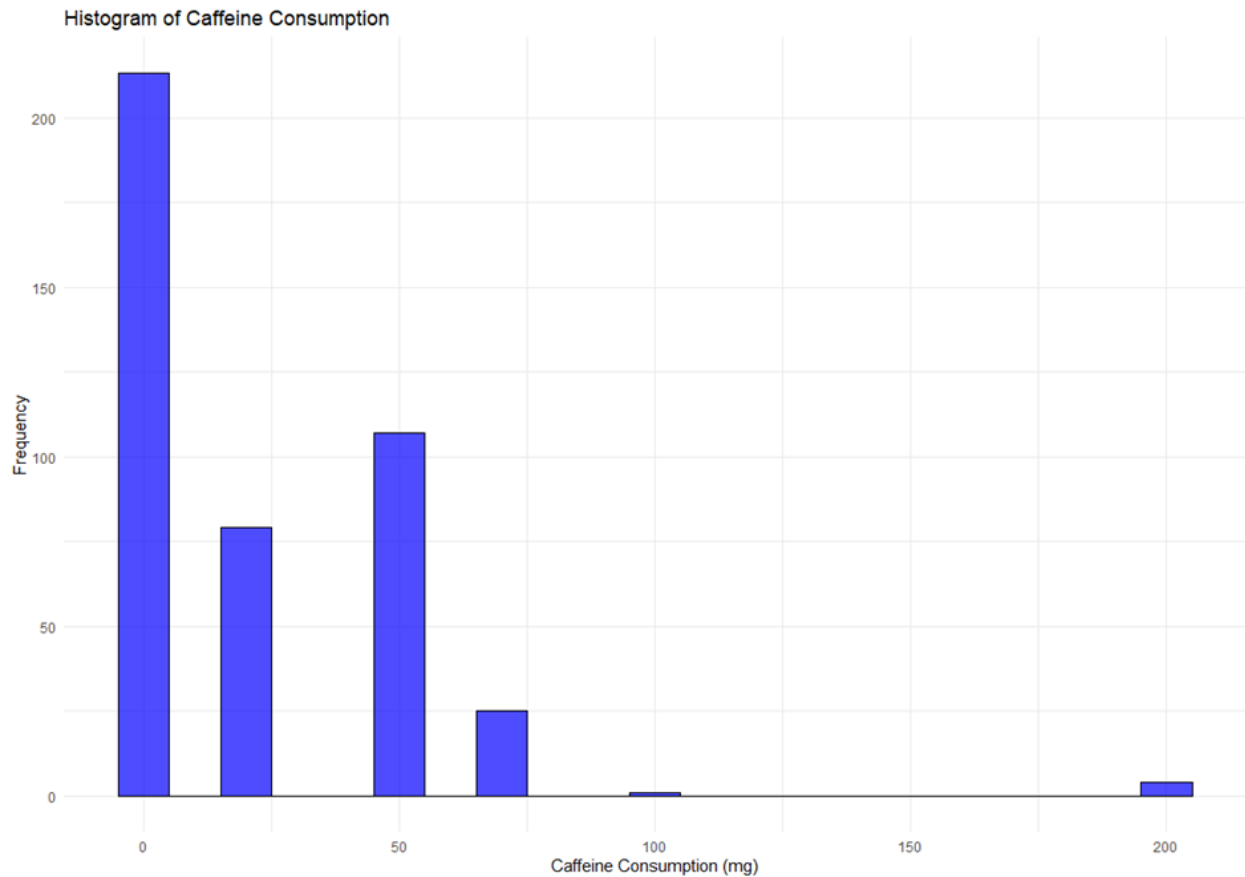


Chart 1(One Sample T-Test)

Chart 1: The histogram above shows that most people on average do not consume any caffeine. There are extreme outliers, where a few people have consumed 200mg of caffeine. The highest population that has consumed caffeine is 50mg.

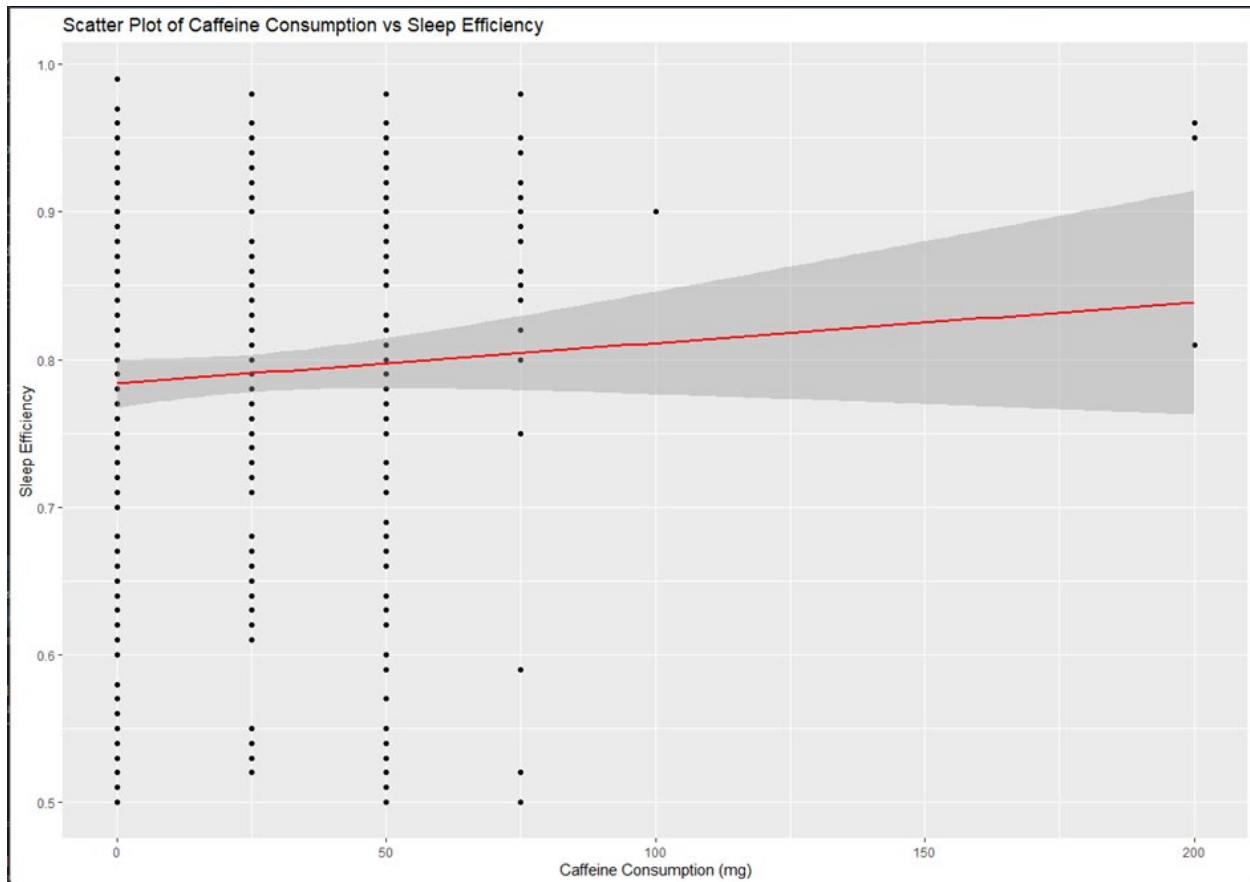


Figure 1 (Correlation)

Figure 1: The scatter plot above shows that there is no correlation between caffeine consumption and sleep efficiency. The average shows that most people that consumed caffeine usually still have an average of 0.8+ sleep efficiency. Since the track line is on an upwards trajectory.

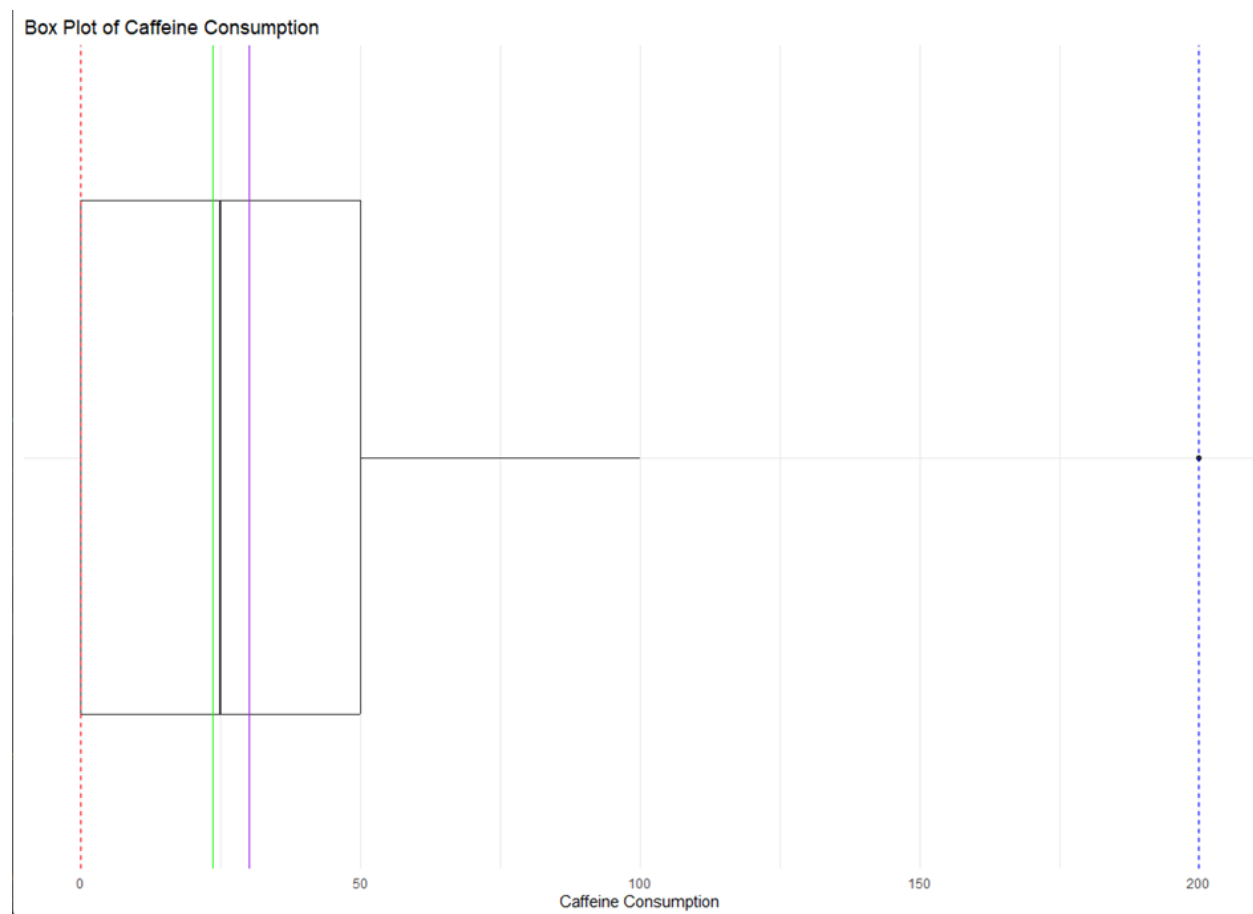


Figure 2 (Descriptive)

Figure 2 Interpretation: The box plot indicates that most individuals have a caffeine consumption between 0 and 50 mg. The mean caffeine consumption is approximately 23.54 mg, with a standard deviation of 30.18 mg, suggesting high variability around the mean. There are no extreme outliers.

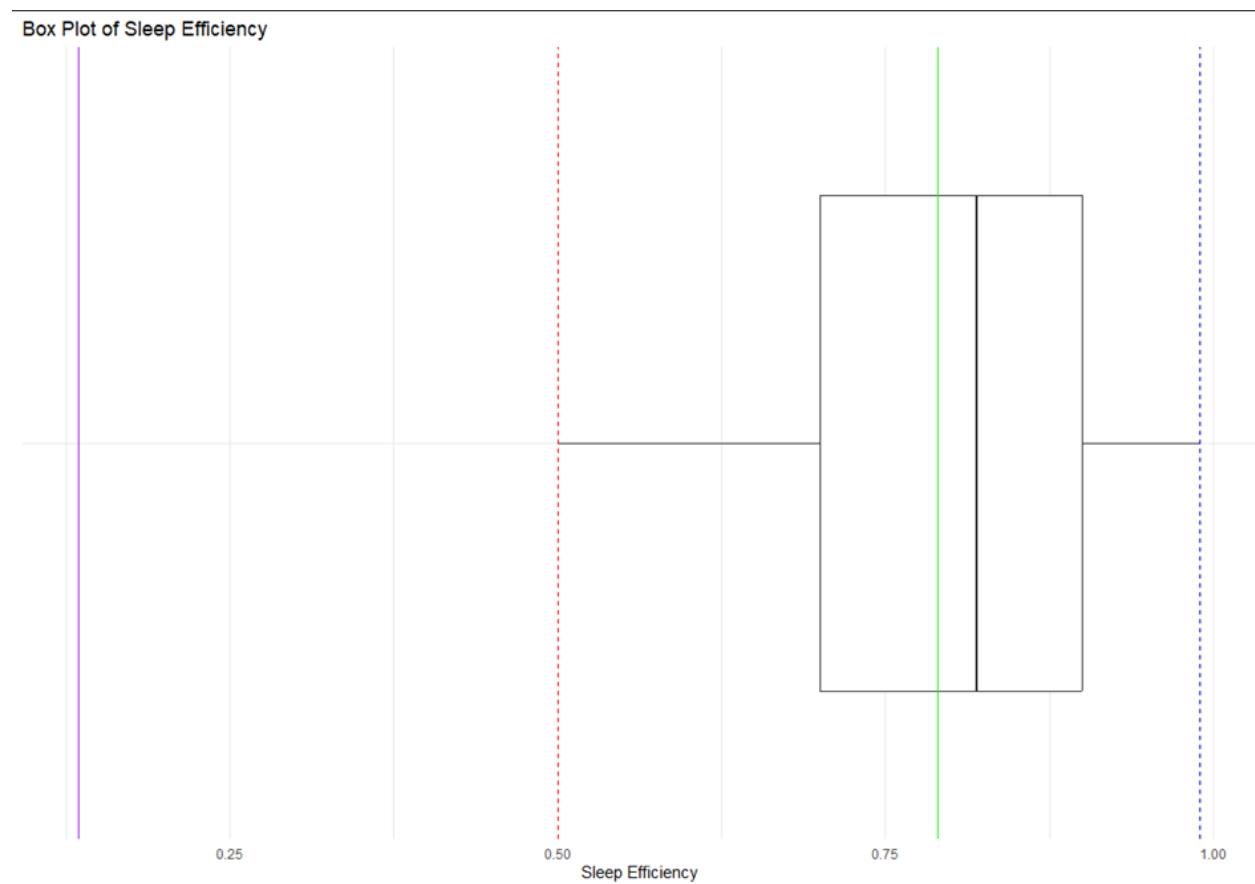


Figure 3(Descriptive)

Figure 3 Interpretation: The box plot indicates that most individuals have a sleep efficiency between 0.66 and 0.88. The mean sleep efficiency is approximately 0.79, with a standard deviation of 0.13, suggesting moderate variability around the mean. There are no extreme outliers.

3.1 Interpret the results (2 Interpretations)

The results suggest that caffeine consumption does not significantly impact sleep efficiency. Although there is a wide range of caffeine consumption among individuals, it does not seem to affect sleep efficiency much

References and Notes

IGB20303/ IGB21303

Carissa Gardiner , Jonathon Weakley , Louise M. Burke , Gregory D. Roach , Charli Sargent , Nirav Maniar , Andrew Townshend , Shona L. Halson. *The effect of caffeine on subsequent sleep*:
<https://www.sciencedirect.com/science/article/pii/S1087079223000205#:~:text=A%20systematic%20search%20of%20the,sleep%20onset%20of%2012%20min.>

R Code for Data Analysis

1) Descriptive

Mean and Standard Deviation

```
## Mean and Standard Deviation
```

```
# Load necessary libraries
```

```
library(ggplot2) # For creating plots
```

```
# Read the CSV file
```

```
data <- read.csv("C:/Users/party/Downloads/Sleep_Efficiency.csv", header = TRUE)
```

```
# Calculate mean and standard deviation for Sleep Efficiency
```

```
mean_sleep_efficiency <- mean(data$Sleep.efficiency)
```

```
std_sleep_efficiency <- sd(data$Sleep.efficiency)
```

```
min_sleep_efficiency <- min(data$Sleep.efficiency)
```

```
max_sleep_efficiency <- max(data$Sleep.efficiency)
```

```
# Print the results
```

```
cat("Mean Sleep Efficiency:", mean_sleep_efficiency, "\n")
```

```
cat("Standard Deviation of Sleep Efficiency:", std_sleep_efficiency, "\n")
```

```
# Calculate mean and standard deviation for Caffeine Consumption
```

```
mean_caffeine_consumption <- mean(data$Caffeine.consumption)
```

```
std_caffeine_consumption <- sd(data$Caffeine.consumption)
```

```
min_caffeine_consumption <- min(data$Caffeine.consumption)
```

```
max_caffeine_consumption <- max(data$Caffeine.consumption)
```

```
# Print the results
```

```
cat("Mean Caffeine Consumption:", mean_caffeine_consumption, "\n")
```

```
cat("Standard Deviation of Caffeine Consumption:", std_caffeine_consumption, "\n")
```

```
# Create a horizontal box plot for Sleep Efficiency with mean and standard deviation  
lines
```

```
ggplot(data, aes(x = Sleep.efficiency, y = "")) + # Using y = "" to create a single category  
on the y-axis
```

```
  geom_boxplot() + # Add the box plot
```

```
  geom_vline(xintercept = min_sleep_efficiency, linetype = "dashed", color = "red") + #  
Add line for minimum value
```

```

    geom_vline(xintercept = max_sleep_efficiency, linetype = "dashed", color = "blue") + #
Add line for maximum value
    geom_vline(xintercept = mean_sleep_efficiency, linetype = "solid", color = "green") + #
Add line for mean value
    geom_vline(xintercept = std_sleep_efficiency, linetype = "solid", color = "purple") + #
Add line for standard deviation
    labs(title = "Box Plot of Sleep Efficiency", x = "Sleep Efficiency") + # Add title and x-
axis label
    theme_minimal() + # Use a minimal theme for the plot
    theme(axis.title.y = element_blank(), # Remove y-axis title
          axis.text.y = element_blank(), # Remove y-axis text
          axis.ticks.y = element_blank(), # Remove y-axis ticks
          axis.line.y = element_blank()) # Remove y-axis line

# Create a horizontal box plot for Caffeine Consumption with mean and standard
deviation lines
ggplot(data, aes(x = Caffeine.consumption, y = "")) + # Using y = "" to create a single
category on the y-axis
    geom_boxplot() + # Add the box plot
    geom_vline(xintercept = min_caffeine_consumption, linetype = "dashed", color = "red")
+ # Add line for minimum value
    geom_vline(xintercept = max_caffeine_consumption, linetype = "dashed", color =
"blue") + # Add line for maximum value
    geom_vline(xintercept = mean_caffeine_consumption, linetype = "solid", color =
"green") + # Add line for mean value
    geom_vline(xintercept = std_caffeine_consumption, linetype = "solid", color = "purple")
+ # Add line for standard deviation
    labs(title = "Box Plot of Caffeine Consumption", x = "Caffeine Consumption") + # Add
title and x-axis label
    theme_minimal() + # Use a minimal theme for the plot
    theme(axis.title.y = element_blank(), # Remove y-axis title
          axis.text.y = element_blank(), # Remove y-axis text
          axis.ticks.y = element_blank(), # Remove y-axis ticks
          axis.line.y = element_blank()) # Remove y-axis line

```

2) Hypothesis Test

One-Sample T-Test

```
## one sample t test
library(ggplot2)
data <- read.csv("C:/Users/khair/Downloads/Sleep_Efficiency.csv", header = T )
caffeine_consumption <- data[,1]

#The daily recommendation of caffeine consumption is 400mg
#population standard deviation is unknown.
#Samples shows the amount of sample caffeine consumption
#Use 95% confidence level

#H0: mean is equal to 400mg
#H1: mean is not equal to 400mg

t.test(caffeine_consumption, mu = 400)
# p value = 2.2e-16 < 0.05, we REJECT H0
# so, we can conclude that the mean of caffeine consumption is not equal to 400

ggplot(data, aes(x = caffeine_consumption)) +
  geom_histogram(binwidth = 10, fill = "blue", color = "black", alpha = 0.7) +
  labs(title = "Histogram of Caffeine Consumption",
       x = "Caffeine Consumption (mg)",
       y = "Frequency") +
  theme_minimal()
```

3) Correlation or Regression

Pearson Correlation Coefficient

```
##PROJECT
library(ggplot2)
#Calling the data
data <- read.csv("C:/Users/khair/Downloads/Sleep_Efficiency.csv", header = T)
x <- data[,1] # To assign x as column 1
y <- data[,2] #To assign y as column 2

#Pearson Correlation
cor(x, y, method = "pearson") #Or you can use this
```

#0.0614591 is close to 0. Therefore, weak positive correlation

#But, we can't conclude for the whole population. so we do this:

H0: There is no correlation between caffeine consumption and sleep efficiency.

H1: There is a correlation between caffeine consumption and sleep efficiency.

cor.test(x, y, method = "pearson")

p-value = 0.2039 > 0.05, Fail to reject H0.

Thus, there is no correlation between caffeine consumption and sleep efficiency.

0.06 indicates that there is a weak correlation between IQ and math.

```
ggplot(data, aes(x = Caffeine.consumption, y = Sleep.efficiency)) +  
  geom_point() +  
  geom_smooth(method = 'lm', col = 'red') +  
  labs(title = 'Scatter Plot of Caffeine Consumption vs Sleep Efficiency',  
        x = 'Caffeine Consumption (mg)', y = 'Sleep Efficiency')
```